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CMSC 471 Section 02

Question 1:

Steps:

1. Step's heuristic formula is simply the number of letters that differ between the current state and the goal state.
2. The unavoidable remaining cost it lower bounds is the ideal scenario where every letter corrects one incorrect letter on current state without changing the letter into another incorrect letter.
3. The lowest cost possibly obtainable is equal to the amount of letters that differ. At minimum you would need to spend one move switching each of these letters at least once. Thus, there are no solutions that would result in a lower cost than the amount of letters.

Scrabble:

1. The heuristic is determined by comparing the letters in each position of the goal state with the corresponding letters of the current state. The letter at the first position of the goal state and the first position of the current state get compared, for instance. This repeats until the end, and if the letters between the two positions differ, the scrabble cost of the goal state's letter at that position is added. The total scrabble value of all differing goal letters is the value that is returned as the heuristic
2. The lower bound represented by this heuristic is the scenario where each letter change results in one letter being converted into a correct letter, happening until each of the letters in the current state matches the goal state.
3. Regardless of the letter's costs, each of the incorrect letters will need to be converted into the correct letters of the goal state eventually. At minimum it will be the sum of the scrabble values for each differing letter in the goal state.

Frequency.

1. The heuristic is equal to the number of letters that differ between the current state and the goal state plus the lowest frequency value among every word that is legally reachable by the current state.

2. The lower bound represented by this heuristic is a scenario where the lowest cost legal move is undertaken at the current state, and every letter change only costs the unavoidable cost of 1.
3. Because the heuristic is equal to the lowest frequency word that can legally be taken plus the number of differing positions, there is no legal move that leads to a cost that is less than the heuristic, meaning it cannot overestimate the true cost. Even if the lowest-cost legal move does not lead to the optimal path, the word used to get to the optimal path will be at minimum the 1 plus its frequency (Which was already established to be greater than or equal to the lowest-cost legal move) plus any other moves necessary (which number at minimum, the amount of differing positions) to reach the goal state if it does not immediately reach it.

Question 2:

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost.	A* nodes expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	765	9	13	97.91%
WHY → ASK	Frequency	38.85540652	808	38.85540652	642	20.54%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402284	1084	38.2402284	483	55.44

1. Yes they always returned the same optimal cost.

2. For Why/Ask, Steps produced the largest percentage reduction. For Bear/duck it was Scrabble.
3. For both the word pairs, the heuristic was least informative under frequency.
4. Both heuristics for steps and scrabble consider the number of incorrect positions, though steps treats each move equally as a cost of 1, making it more accurate measure moves remaining. Scrabble on the other hand assigns values for each letter making it sensitive to the specific letters used.

The frequency heuristic only considers the cheapest frequency word that can be legally reached from the current state and the number of incorrect positions, not accounting for frequency costs of words encountered later in the path or other paths from the current state, making it significantly less informative than the other two heuristics.

Question 3:

Steps

1. Yes.
2. Because all moves cost 1, every move and their reverse will always have equal cost. Because of this the optimal path from word 1 to word 2 can be reversed to produce an optimal path from word 2 to word 1, which results in their optimal costs being the same.

Scrabble

1. No.
2. Cost depends on the letter. Thus. If the difference between two words is a singular letter, and changing that singular letter happens to the optimal path for word 1 transitioning to word 2 and vice versa, that would mean that the optimal cost could differ as long as those two letters have two different scrabble values.
3. Take log and dog.

L scrabble value = 1

D scrabble value = 2

Log->Dog optimal path is replacing L with D.

Dog->Log optimal path is replacing D with L.

Log->Dog cost = 2.

Dog->Log cost = 1

$1 \neq 2$, thus optimal cost differs.

Frequency

1. No,
2. Transitioning from one word to another gives a certain cost. But the transition back into the previous letter is not guaranteed to cost the same. Based on this principle, imagine there is word 1 and word 2 with a frequency of f_1 and f_2 respectively where $f_1 \neq f_2$. Now let us suppose then that the most optimal path involves a single optimal move for the transition from word 1 and word 2 is to change a single letter, which results word 1 reaching to goal state (word 2). The cost for this single move is $1 + f_2$. For the reverse, where word 2 is the initial state and word 1 is the goal state, suppose that the most optimal path is again a singular move that results in the goal state (word 1). The cost for this move would be $1 + f_1$. Both optimal moves are not equivalent as $1 + f_1 \neq 1 + f_2$ because $f_1 \neq f_2$.
3. Can frequency: 1.47
Man frequency: 2.65

Optimal path Can->Man, Replace C with M.

Optimal path, Man->Can, Replace M with C.

Can->Man optimal cost = $2.65 + 1 = 3.65$

Man->Can optimal cost = $1.47 + 1 = 2.47$

$3.65 \neq 2.47$ Thus optimal cost differs.

Question 4:

1. The model used was Google Gemini 3.5 Flash-Lite. The exact prompt given was the stated prompt provided in the instruction pdf.
- 2.

Proposal for Rarity-Aware Heuristic (h_1)

To design a more informative yet admissible heuristic, we can incorporate the global minimum rarity of the dictionary while respecting the structural lower bound established by the baseline heuristic $h_0(n)$.

Definition

Let $R_{\min} = \min_{w \in \text{Dict}} \text{rarity}(w)$ represent the minimum rarity value across all words in the dictionary. We define the rarity-aware heuristic $h_1(n)$ as:

$$h_1(n) = h_0(n) \cdot (1 + R_{\min})$$

where $h_0(n)$ is the number of positions in which state n differs from the goal g (Hamming distance).

3. This heuristic is admissible. The frequency value is guaranteed to be the lowest globally. Thus part of the heuristic $(1 + R_{\min})$ represents the absolutely minimum cost that any move could be with a given dictionary. Because of this, every legal move must be either equal to or greater than the absolute minimum value. And since this minimum move is multiplied by how many differing positions there are this results in the globally minimum cost path that could ever be possible to reach the goal state for a given state. Thus, no path can lead to a lesser value than the given heuristic.
4. N/A
- 5.

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost.	A* nodes expanded	Reduction
WHY → ASK	Frequency	38.86	808	38.86	754	6.68%
BEAR → DUCK	Frequency	38.24	1084	38.24	714	34.13%

The Heuristic led to the same cost as UCS which provides some evidence that it is admissible. The percent reduction varied wildly, Why->Ask Yielded only a 6.68% reduction but Bear->Duck yielded a 34.13% reduction. While the reduction amount varies, it seems to reduce the amount of nodes by a low to moderate amount.